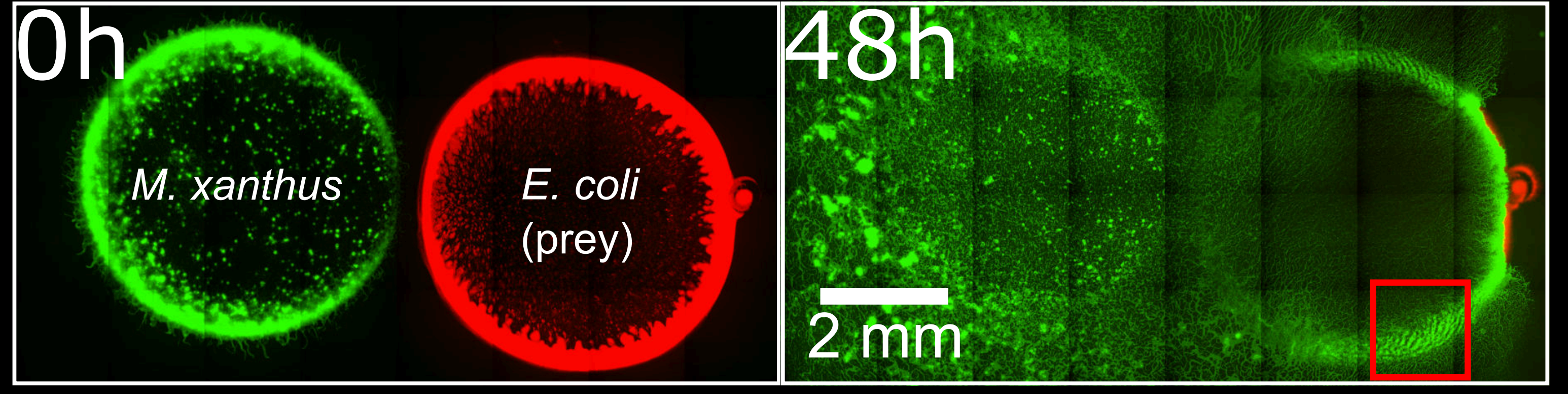
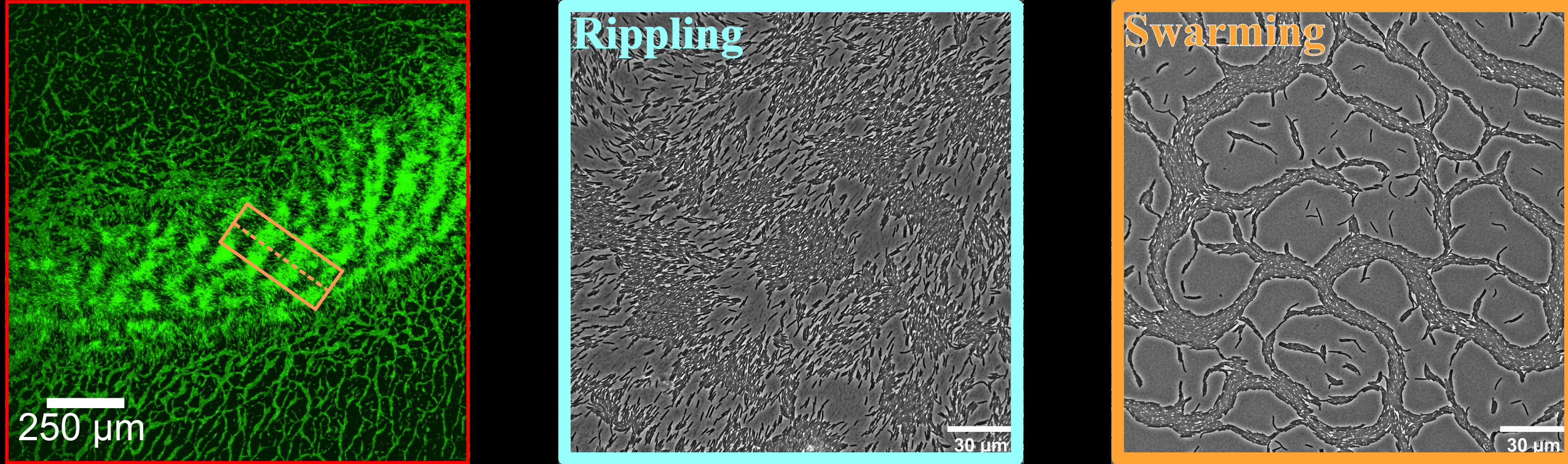


To understand how individual behaviours contribute to collective actions across scales-ranging from molecular interactions to animal populations-it is essential to identify the fundamental rules that govern these interactions. *Myxococcus xanthus*, a bacterial predator, can produce very large multicellular patterns, transitioning from a swarming behaviour to a rippling behaviour due to changes in the local environment within and around prey colonies. Through a combination of high-resolution microscopy and theoretical analysis, we demonstrate that this transition can be explained by two simple properties, local cellular alignment guided by an extracellular matrix and the ability of cells to resolve congestions by reversing. Furthermore, we show that a tunable refractory period in the reversal control system enables a wide range of collective adaptations, allowing cells to synchronise during rippling and relieve congestions during swarming. Remarkably, these transitions occur without genetic regulation changes, yet they result in stable spatial domains that facilitate local differentiation.



Rippling
Swarming

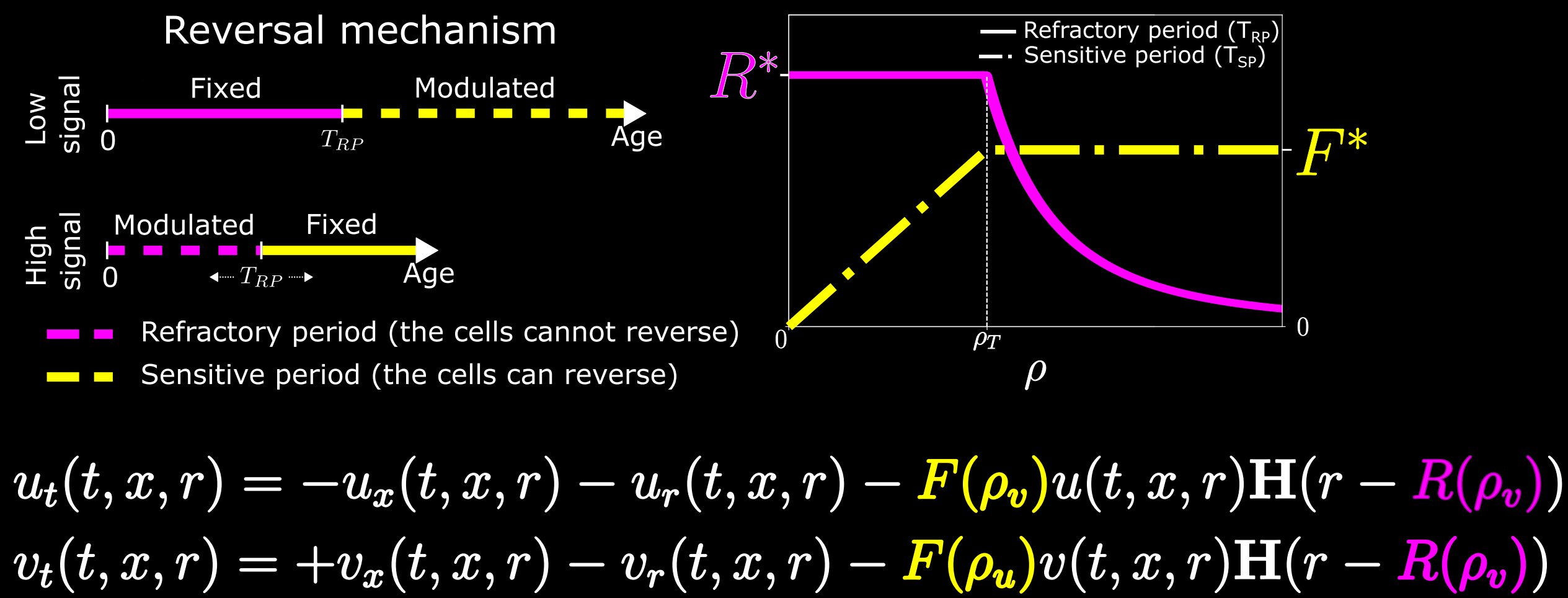
1. Introduction and defining the transition



Observations show that rippling (wave-like structures) and swarming (mesh-like structures) coexist through a sharp transition. The rippling pattern is characterized by a highly aligned structure and a bimodal time distribution of reversals (TBR), whereas swarming exhibits weak alignment beyond a few cell lengths and displays a unimodal TBR distribution (see Sec. 2).

What mechanisms underlie the transition between rippling and swarming?

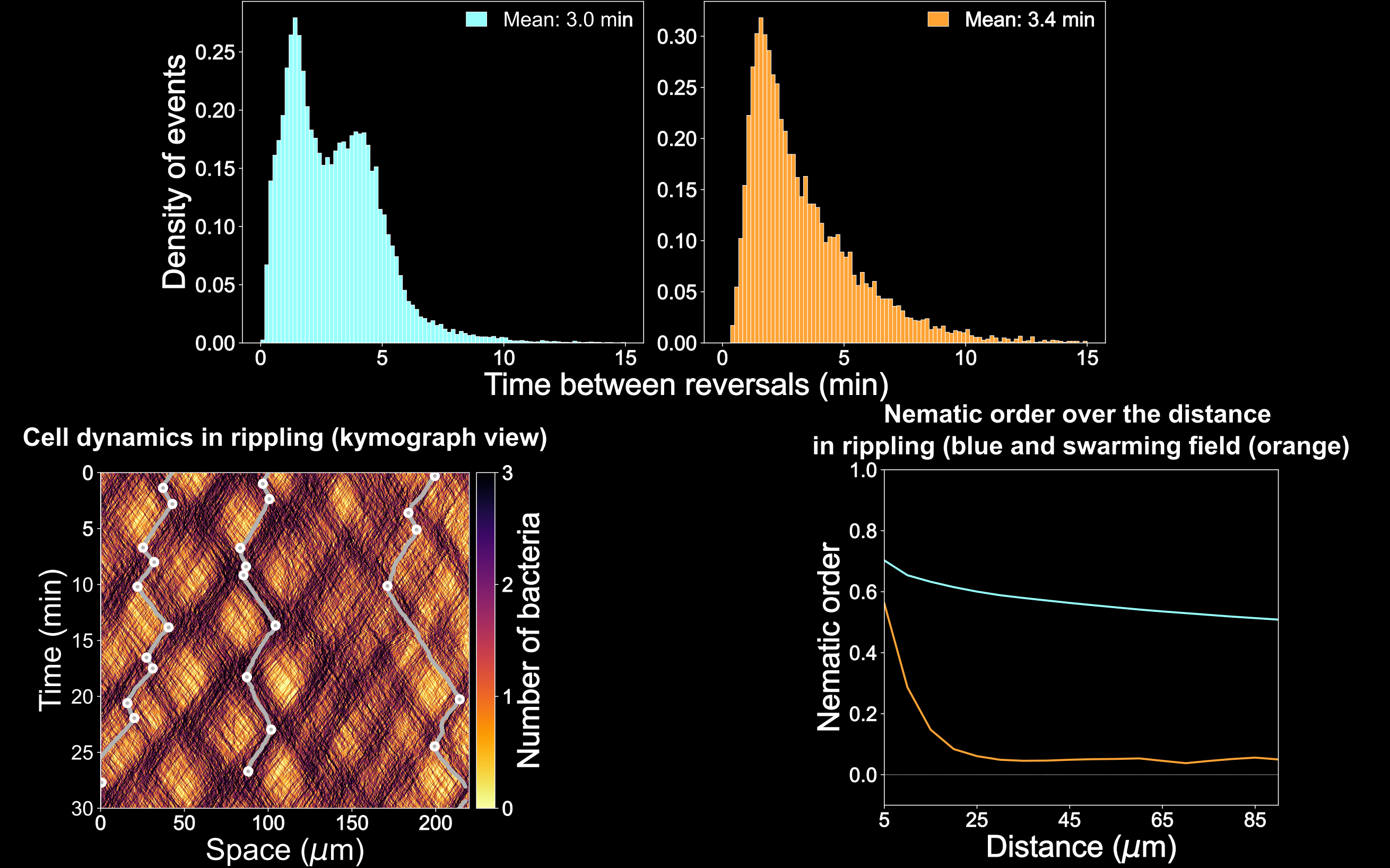
3. 1D simulation of rippling and instability analysis



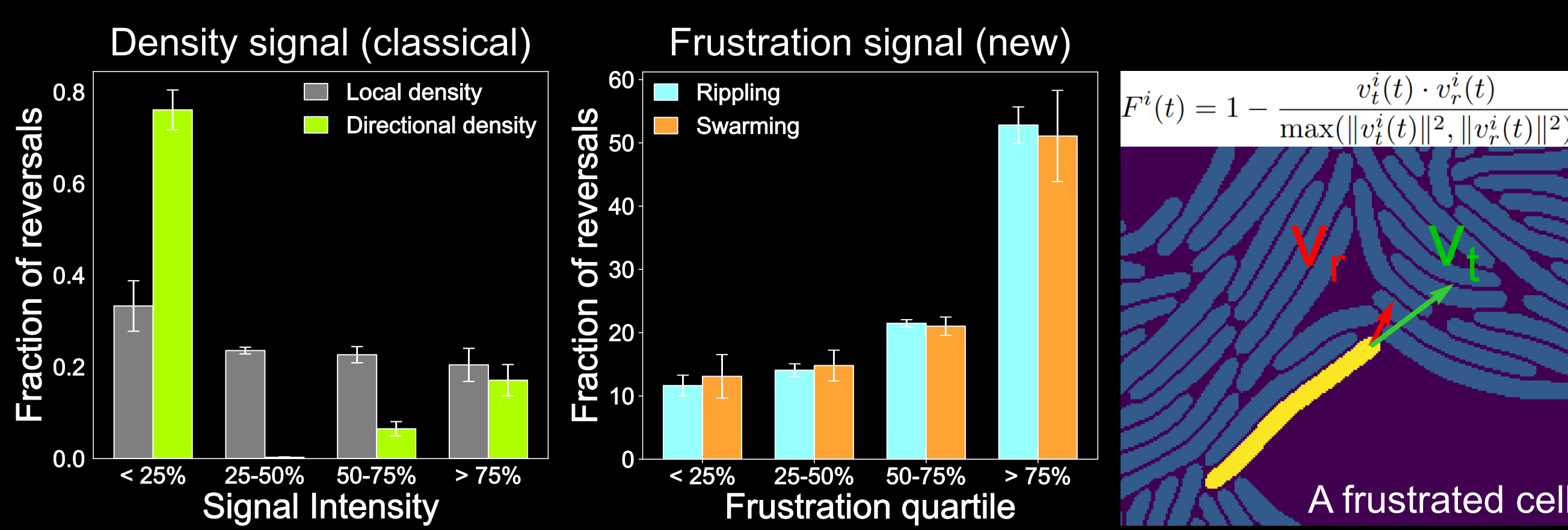
$$u_t(t, x, r) = -u_x(t, x, r) - u_r(t, x, r) - F(\rho_v)u(t, x, r)H(r - R(\rho_v))$$

$$v_t(t, x, r) = +v_x(t, x, r) - v_r(t, x, r) - F(\rho_u)v(t, x, r)H(r - R(\rho_v))$$

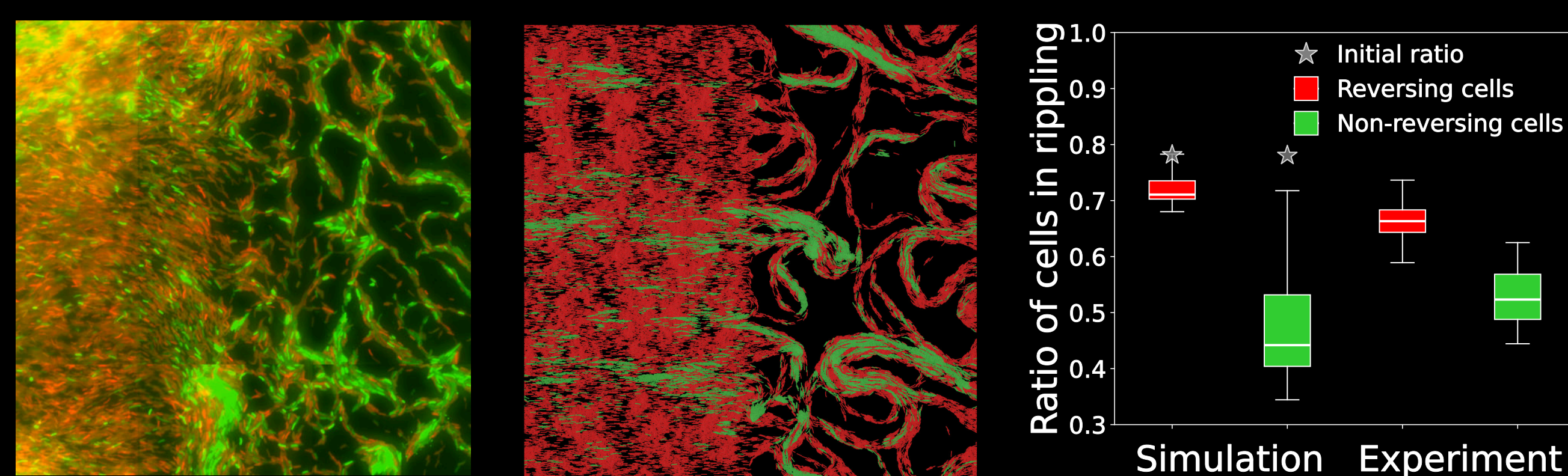
2. Experimental metrics to define rippling and swarming



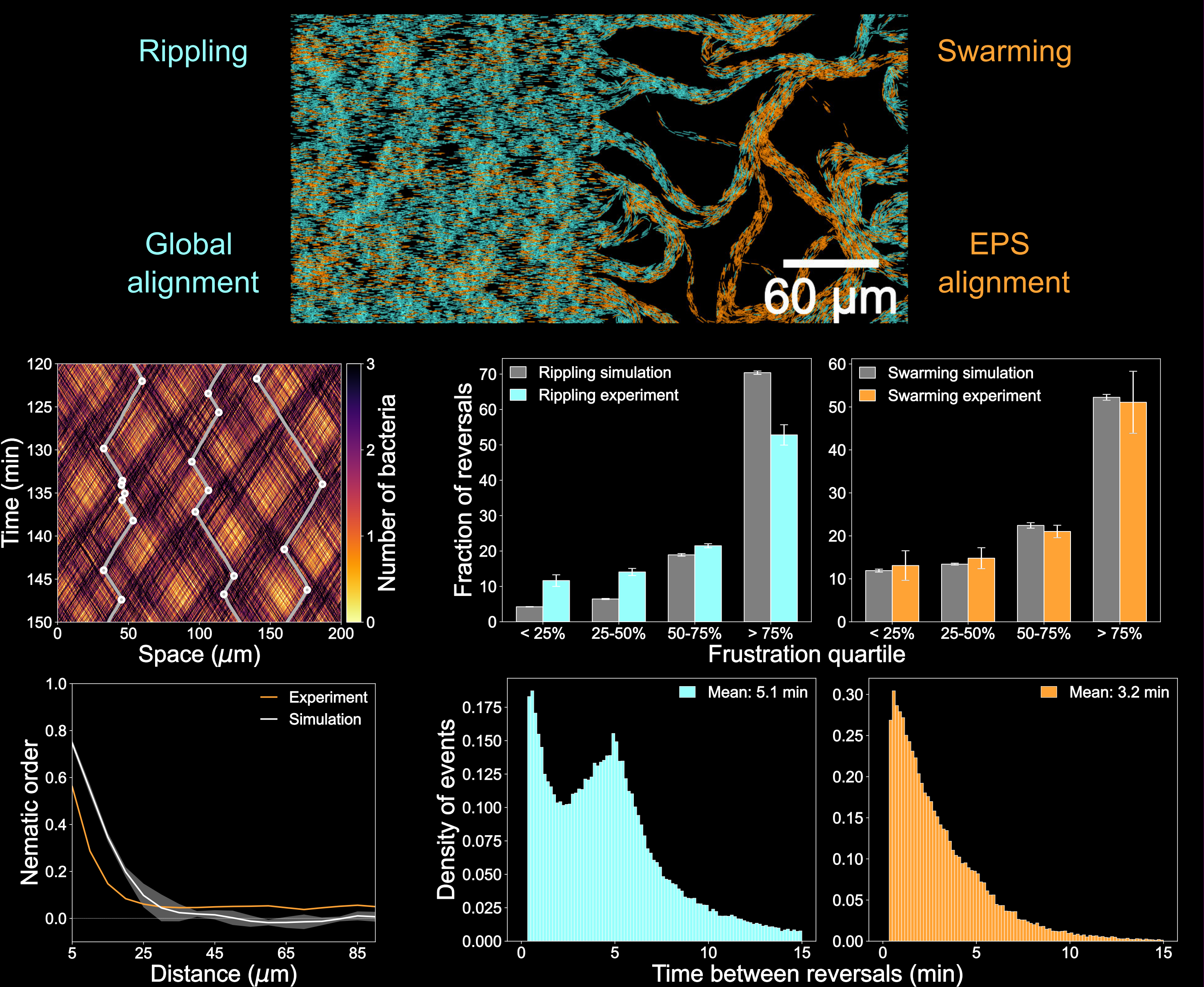
4. Density versus frustration reversal signaling



6. Rippling has a cell trapping effect mechanism



5. 2D agent based simulation with integrated frustration



7. Conclusion

We have demonstrated the importance of refractory period modulation in the reversal mechanism (Section 3). Additionally, the classical reversal signal (density) does not align with experimental observations; instead, we introduced a new signal, called **frustration**, which matches experimental results (Section 4). Furthermore, using a **2D agent-based model**, we have shown that the emergence of the **rippling-swarming transition** can be explained by differences in **cell alignment**—with **global alignment** leading to rippling and **EPS cell following** leading to swarming but keeping the same reversal regulation through the frustration signal (Section 5). Finally, we have identified the **trapping effect** of the rippling phenomenon, which allows cells to regulate their reversals periodically. This effect was experimentally validated, demonstrating the same behavior as observed in simulations (Section 6).